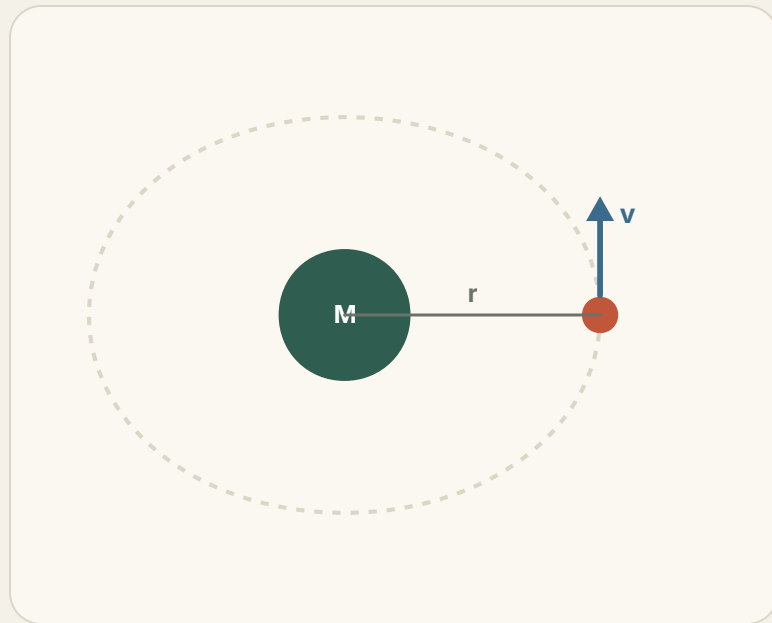


Achievements of Gravitation

Newton's law became a tool for weighing worlds and finding bodies nobody had seen.

ESSENTIAL QUESTION

How can motion near one body reveal the mass, density, and even the existence of another?



● LESSON MAP

Turn one law into four tools.

01

WEIGH EARTH

Use surface gravity once
G is known.

02

WEIGH STARS

Use an orbiting body as a
moving probe.

03

FIND DENSITY

Combine mass with radius,
or use a low orbit.

04

PREDICT

Explain orbit errors and
forecast returns.

THE COMMON IDEA

A body in circular motion is a measuring instrument: its orbit reveals the **pull** that guides it.

THE PROBLEM

A planet cannot be put on a scale.

A balance compares two objects. Earth is what makes the scale work, so it cannot weigh itself.

1 Direct weighing fails

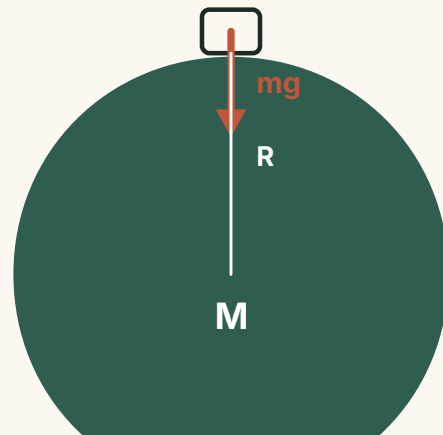
A balance would still need something outside Earth to compare against.

2 Gravity gives an indirect route

Know the force law, and nearby motion reveals the mass making the field.

3 Cavendish supplied the number

Once G was measured, the indirect route became a calculation.



METHOD 1

Surface gravity weighs the central body.

A falling object helps measure the planet — yet its own mass disappears.

WEIGHT IS GRAVITY

$$mg = G \frac{Mm}{R^2}$$

CANCEL THE TEST MASS

$$M = \frac{gR^2}{G}$$

The unknown is the planet's mass M .



WORKED RESULT

Cavendish made Earth's mass a number.

Use the values measured near Earth: $g \approx 9.8 \text{ m s}^{-2}$, $R \approx 6.4 \times 10^6 \text{ m}$, and G .

$$M = \frac{gR^2}{G}$$

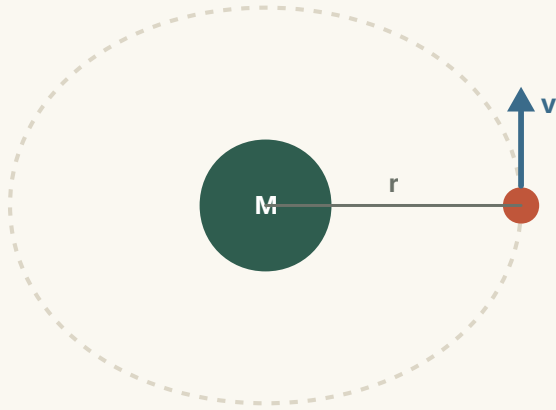
CALCULATION

Once G was measured, the formula could be used — giving $M_{\oplus} \approx 6 \times 10^{24} \text{ kg}$.

METHOD 2

An orbiting body weighs what it orbits.

For a near-circular orbit, gravity supplies the centripetal force; the orbiting mass cancels.



FORCE BALANCE

$$G \frac{Mm}{r^2} = m \frac{4\pi^2 r}{T^2}$$

SOLVE FOR THE CENTRE

$$M = \frac{4\pi^2 r^3}{GT^2}$$

The orbit gives the central mass, not the satellite's.

WORKED RESULT

Earth's year weighs the Sun.

Treat Earth's orbit as circular with $r \approx 1.5 \times 10^{11}$ m and $T \approx 1$ year.

$$M = \frac{4\pi^2 r^3}{GT^2}$$

ORBITAL CALCULATION

The planet is the probe; the Sun is the unknown central mass — $M_{\odot} \approx 2 \times 10^{30}$ kg.

DENSITY

Mass plus radius gives average density.

Density is bookkeeping after a mass has been found — not a new force idea.

FROM MASS AND RADIUS

$$\rho = \frac{M}{\frac{4}{3}\pi R^3}$$

USING SURFACE GRAVITY

$$\rho = \frac{3g}{4\pi GR}$$

A near-surface shortcut.

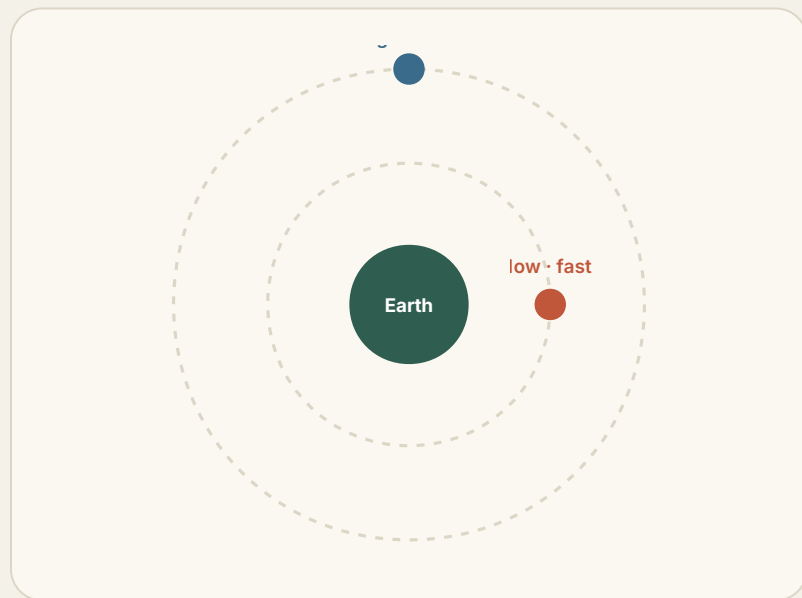
WHEN TO USE WHICH

Use surface gravity and radius for a surface measurement; use orbital period and radius when an orbiting body is available.

ORBIT COMPARISON

Higher circular orbits move more slowly.

For satellites around the same body, the orbital radius controls every motion parameter.



ORBITAL SPEED

$$v = \sqrt{\frac{GM}{r}}$$

ORBITAL PERIOD

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

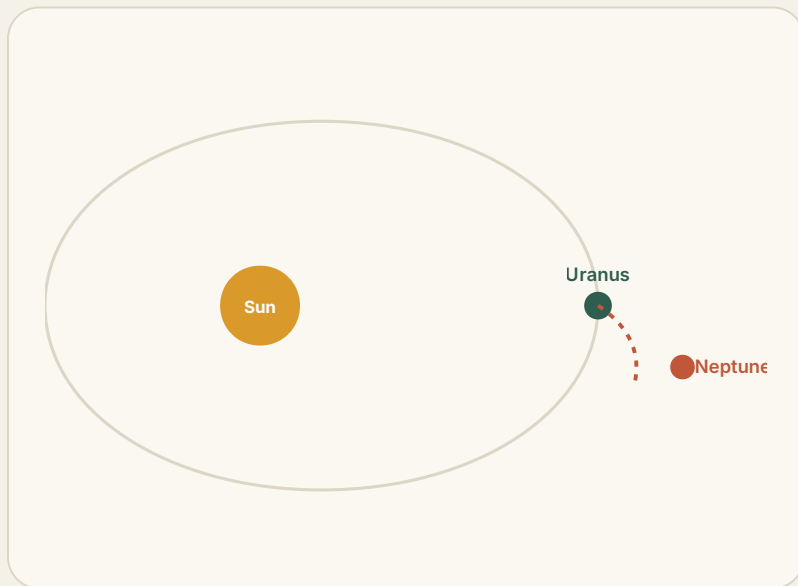
Higher orbit → lower speed, longer period.

PREDICTION

Orbit errors can reveal unseen worlds.

Uranus did not quite follow the path computed from the known planets.

- 1 Observation disagreed**
Small residuals in Uranus's position hinted at an extra pull.
- 2 Mathematics pointed the telescope**
Adams and Le Verrier estimated where another planet must be.
- 3 Neptune was found in 1846**
Galle saw it near the predicted spot — a triumph of Newtonian gravity.



PREDICTION

A comet's return can be forecast.

Halley realised several recorded comets were one object on a long elliptical orbit.

1 An orbit, not just a force

Once the path is known, the period can be estimated.

2 Halley predicted a return

The comet came back in 1758 — after his death, as predicted.

3 Next return: 2061

The law that weighs planets also forecasts the sky.

THE PATTERN

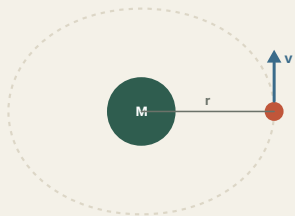
Motion is evidence for the force; the force is evidence for mass. Gravity turned astronomy into measurement.

● SUMMARY

Gravity turned astronomy into **measurement**.

WEIGH THE CENTRE FROM AN ORBIT

$$M = \frac{4\pi^2 r^3}{GT^2}$$



Surface gravity weighs a body; an orbit weighs what it circles; mass and radius give density.

Small deviations point to unseen pulls — and known orbits forecast the future.